A MARL Framework for Adaptive Differential Privacy in Byzantine-Resilient Federated Learning

by

Afnan Satter
22201765
Rabsa Anjum Prosongsha
24241347
Masrura Nahian Nodi
22201421
Mohsina Binte Harun
24301586
Tasnim Nida
22201954

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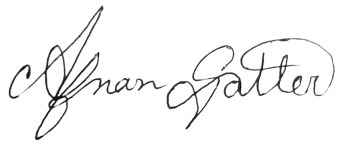
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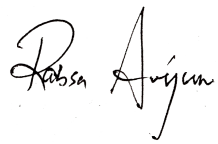
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Student's Full Name & Signature:



Afnan Satter
22201765



Rabsa Anjum Prosongsha
24241347



Mohsina Binte Harun
24301586



Masrura Nahian Nodi
22201421



Tasnim Nida
22201954

Approval

The thesis/project titled " A MARL Framework for Adaptive Differential Privacy in Byzantine-Resilient Federated Learning" submitted by

1. Afnan Satter (22201765)
2. Rabsa Anjum Prosongsha (24241347)
3. Masrura Nahian Nodi (22201421)
4. Mohsina Binte Harun (24301586)
5. Tasnim Nida (22201954)

of Fall, 2025 has been accepted as satisfactory in partial fulfillment of the requirement for the degree of B.Sc. in Computer Science in Fall 2025.

Examining Committee:

Supervisor:
(Member)



Dr. Muhammad Iqbal Hossain

Associate Professor, CSE
Department of Computer Science and Engineering
Brac University

Thesis Coordinator:
(Member)

Dr. Md. Golam Rabiul Alam

Professor
Department of Computer Science and Engineering
Brac University

Head of Department:
(Chair)

Dr. Sadia Hamid Kazi

Chairperson
Department of Computer Science and Engineering
Brac University

Abstract

While Federated Learning (FL) enables decentralized model training, the presence of malicious participants who submit corrupted or poisoned updates poses a severe threat to model convergence and integrity. Traditional aggregation methods often struggle when applied in decentralized model training with RL tuned Differential Privacy (DP) noise, as static noise injection typically degrades model utility or fails to account for the dynamic nature of adversarial attacks. To address this, we propose a novel Multi Agent Reinforcement Learning (MARL) framework for Adaptive Privacy Budget allocation in Byzantine resilient FL environments. In this framework, multiple RL agents collaborate to dynamically calibrate DP noise levels based on the observed variance and reliability of local updates. By learning the optimal balance between privacy-preserving noise and model utility, the MARL controller effectively masks sensitive data while isolating anomalous Byzantine contributions. Furthermore, our mechanism utilizes the dynamic noise optimization method (DNO-RL) that dynamically adjusts the Gaussian noise scale, ensuring that noise injection is minimized during critical convergence phases and intensified when potential poisoning is detected. This adaptive approach enables the system to maintain Byzantine resilience without the catastrophic accuracy loss associated with fixed-rate DP. Extensive experiments on large-scale datasets demonstrate that our framework achieves superior robustness against label-flipping and gradient-manipulation attacks, providing a scalable solution for trustworthy, privacy-preserving collaborative learning with provable DP guarantees.

Keywords: Federated Learning, Differential Privacy, Byzantine Resilience, Multi-Agent Reinforcement Learning, Adaptive Noise Injection, Dynamic Noise Optimization

Dedication

Thanks to Almighty Allah for giving us the strength and patience to complete this thesis. We would like to dedicate this thesis to our beloved Supervisor Dr. Muhammad Iqbal Hossain sir, RA Fahmid UL Alam Juboraj for unwearied support, encouragement, and understanding throughout the academic journey. Their unwavering belief in us and our dedication and our believe in us has been a constant source of motivation.

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Chapter 1

Introduction

1.1 Background

In current machine-learning research, the terms privacy, security, and trust are distinguished as important preconditions for the increased adoption of information-based intelligence. The paradigms of centralised learning require the close exposure and the ultimate accessibility of datasets of users, which promotes the problem of data exfiltration, compliance with regulations, and exposing the vulnerability of one point of system crash to failure.

Federated learning (FL) is a messaging decentralised alternative that allows optimising the models of a distributed population of clients with the use of only in-device raw data. This architecture removes most of the privacy sensitivity issues present in centralised designs, which makes FL especially appropriate to massively deployed applications with privacy challenges. The decentralised nature gives it strength against a special group of operating risks.

However, FL also comes with its own set of challenges, the first of them being the unification of the model updates that were generated by heterogeneous and, possibly, unreliable participants. Poisoned gradients (an example of the Byzantine behaviours) and rebranding, as well as other malicious software updates, undermine the convergence path and can adversely affect overall model performance, particularly in a situation where traditional aggregation mechanisms are based on the assumption of honest behaviour.

FL deployments are known to often use the mechanisms of differential privacy (DP) to further strengthen privacy assurances. The protective effect of a DP scheme is, however, subject to its particular implementation properties- the noise level, sensitivity estimation, and query budget optimality all have a bearing on the protection utility balance achieved.

Also, fixed privacy budgets drive a strict tradeoff between confidentiality and utility, neither of which is optimally beneficial in a setting where adversarial forces can rise and fall, and clients can be considered reliable or unreliable. The synergising existence of non-homogeneous levels of threats and exploitative behaviour, therefore, further intensifies the challenges described above during the course of the training lifecycle.

1.2 Rational of the Study or Motivation

The primary reason for conducting this study is to resolve the increasing conflict between maintaining privacy and model strength in collaborative machine learning frameworks. Federated Learning (FL) has become a potential alternative to centralized training because it allows a number of clients to collectively learn a global model without necessarily sharing their personal data. Nevertheless, FL does not entirely eradicate privacy risks. This decentralized environment is not ideal in the real world, where clients can be unreliable and untrustworthy, communication channels can be compromised, and self-interested participants can deliberately post updated models that degrade or corrupt model behavior. In the absence of other privacy protection mechanisms, FL is prone to gradient-based privacy attacks that can reveal sensitive client data. More specifically, federated learning systems that lack Differential Privacy (DP) are vulnerable to privacy leakage attacks. The adversaries can use reconstruction attacks to reconstruct individual training samples of facial images, text fragments, or medical data with high fidelity at a single gradient update of a client. Moreover, the membership inference attacks can help to conclude whether a particular person's data was used to train a model, and the property or attribute inference attacks can make a conclusion about such sensitive features of a person as his/her gender, age, or health condition using updates to the model. These breaches of privacy are in direct conflict with the data protection laws and basic privacy rights, which are implemented by the current legal standards. To alleviate these risks, DP is added to FL to add some noise control to client updates before sending them.

In the majority of federated learning, the central server is expected to be honest but inquisitive, that is, they adhere to the protocol in its true sense, but may seek to acquire sensitive information about the updates they receive whenever feasible. Differential Privacy guarantees that, although the server may view the model weights or gradients, identifiable information is obscured by the injected noise to the extent that it is not possible to reconstruct identifiable data. The fact that the aggregate of many clients updates reduces information leakage further and that the global model gets progressively better with each set of training rounds makes the point of aggregating many clients updates in their turn. Consequently, DP is a key privacy protection tool in federated learning systems. Nevertheless, inasmuch as DP theoretically solves the privacy leakage, it presents a strict privacy-utility tradeoff. The vast majority of current methods use deterministic budgets of privacy and constant noise on all clients and training sessions. This fixed setup frequently results in avoidable diminution of accuracy or poor protection in adversarial situations. The constraint is especially strong in settings that are prone to Byzantine attacks, in which the intensity of attacks, trustworthiness of clients, and the behaviour of participation as a phenomenon evolve dynamically. This leads to ineffective and even counterproductive privacy enforcement of all clients and training rounds as the same, irrespective of any observable behavior or quality of contribution, which is effective in many adaptive systems. Equally, federated learning systems need systems that can dynamically respond to changes in training rather than using fixed,

one-size-fits-all policies. This study is inspired by the necessity of smart and context-sensitive privacy controls that can ensure the privacy of data, but at the same time do not reduce model utility or strength because of the learning-based privacy control. This flexibility is necessary to facilitate federated learning to transition from controlled laboratory settings to trustworthy deployment in real-world, adversarial, and privacy-sensitive environments.

1.3 Problem Statement

The field of Federated Learning (FL) has become more and more popular as a privacy-friendly version of training a centralized machine learning model, allowing collaborative model training without sharing data directly. However, the benefits associated with it, FL systems need to run in highly decentralized and distrusted conflicting settings where participating clients can act unstable and malicious manner. These environments expose the global learning process to Byzantine failures, such as poisoned gradients, manipulated labels, and corrupted updates, which can greatly worsen model convergence and corruption. Other mechanisms that may be applied to protect sensitive information are Differential Privacy (DP), which is usually introduced to FL because of noise injection. Even though formal privacy is guaranteed by DP, in practice, there is a serious trade-off between privacy and model utility in its implementation. The majority of the current FL models assume the use of fixed privacy budgets and fixed noise levels, and use the same level of perturbation on each training round and each client. This hardening method does not take into consideration the dynamism of federated settings, where the reliability of clients, distributions of data, and adversarial behavior change as time goes by. This is more amplified in the Byzantine conducive regions. One way that static DP noise can lead to unnecessary learning either in static convergence phases or is inadequate when adversarial activity is increasing, is by itself. Additionally, conventional approaches to aggregation, like FedAvg, are commonly simple enough not to respect or notice aberrant updates, particularly when the DP noises replace the difference between harmless or criminal actions. Consequently, privacy measures and resiliency plans often conflict, as opposed to coordinating. Though there are separate constituents of privacy preservation, Byzantine resilience, and adaptive optimization, they are largely discussed on a case-by-case basis. The existing studies are silent on the need to have a unifying, dynamic structure capable of cleverly managing the enforcement of privacy and, at the same time, remaining resistant to the forces of the adversarial clients during the training exercise. This lack of adaptive control renders this method infeasible in a real-world, adversarial, and privacy-sensitive context, and the lack of such ability poses a serious gap that needs to be closed to ensure that federated learning can be both trusted and useful at scale.

1.4 Objective

The given framework simplifies the formulated federated learning with differential privacy to a Multi-Agent Reinforcement Learning (MARL) problem, whereby single clients would act as independent agents that learn optimal privacy budget allo-

cations to global performance restraints individually. This design is backed by a dynamic noise calibration adaptive mechanism, with which the clients can adjust the amount of noise propagated in the different training rounds without violating cumulative privacy budget constraints. In order to maintain the integrity of operations in adversarial settings, the system incorporates the Byzantine-resilient aggregation methods that can minimize the effect of malicious or abnormal updates, like the ones that are generated during the label-flipping and the gradient-manipulation attacks. A complex reward system directs the learning paths of the agents in order to match the global model utility with collaborative behaviours that are privacy-constrained. A more detailed assessment in real adversarial conditions on benchmark datasets allows comprehensively investigating the underlying privacy-strength-privacy-robustness and model-accuracy trade-offs and proves the greater effectiveness of this adaptive privacy control system than the more classical static differential privacy settings.

1.5 Methodology in Brief

We are and will be using the MARL framework in this study; each client of the federated learning is characterized as a separate reinforcement-learning agent which learns how to spend its privacy budget, yet all agents coordinate their own actions to the global model.
As a preliminary remark to the detailed description of the workflow, we would like to say that we suppose that the server, on which all the aggregations will occur, is an honest yet inquisitive server. And we also viewed our pipelines as being vulnerable where Local Client agents and server are connected, and hence we assumed that an outlier/attacker may seek to make untargeted attacks to compromise the quality of the central models with the help of some network attack tools such as packet injectors to further increase the DP noise, sign flipping, random noise, etc. And the most typical FedAvg is so naive as to pick up or cancelize these disloyal updates, as it is, though, a mere simple average of all the weights and then aggregates. Accordingly, we required a Byzantine fault client update handler to block these malicious updates.

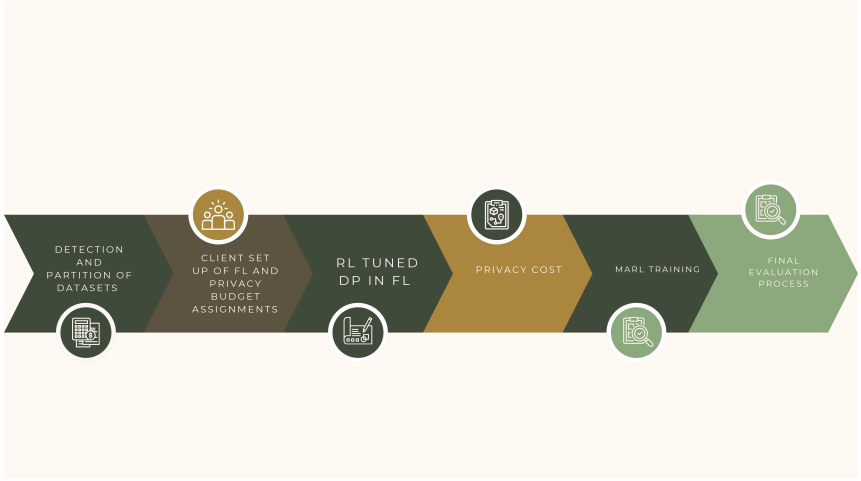


Figure 1.1: Process Flow Diagram

1.5.1 Selection and Partition of Datasets

To begin with, we required some benchmarking datasets that are ideal to play around with in the experimental setting and then apply it to the real world. Therefore, we selected MNIST (handwritten grayscale 28×28 digit images made of 10 classes), and CIFAR-10 (small natural $32 \times 32 \times 3$ RGB images of 10 classes). Where we attempted to tie the IID and Non-IID dataset in which we maintain the balance of the group of clients together so that we can eliminate some forms of bias in our model towards a specific group.

1.5.2 Client Set-up of FL and Privacy Budget Assignment

We initially had ten clients ($K = 10$) to take part in the FL. And we needed to assign a total Privacy budget (ρ_k) per client based on our business requirement, i.e., we were making the assumption that the dataset of some clients is more privileged than that of other clients at random, hence has to get assigned more privacy. In the case of MNIST, ρ_k is uniformly sampled between $[2, 4]$ and in the case of CIFAR-10, we uniformly sampled between $[20, 40]$. Since this dataset is full of colourful images and details, the model will barely be able to learn the right pattern with much noise on it. The clients (k) will each possess a partitioned personal dataset D_k.

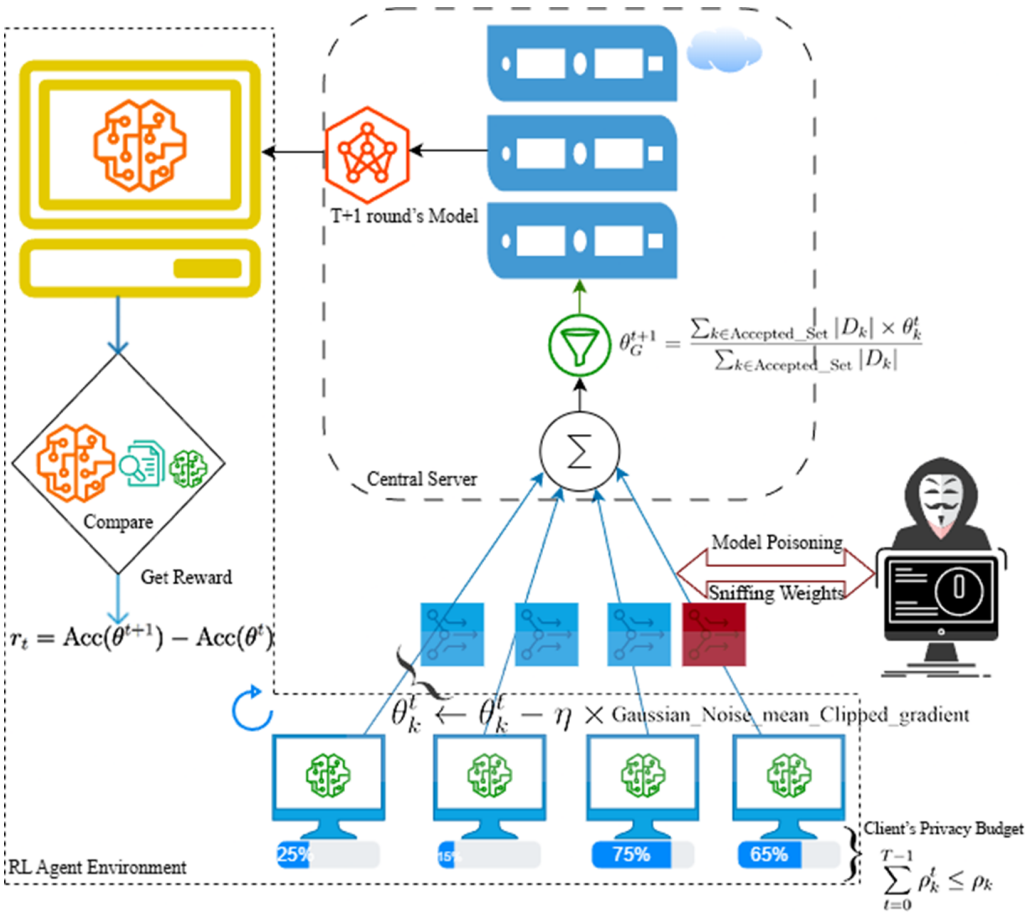


Figure 1.2: Data Pipeline in the Environment

1.5.3 RL Tuned DP in FL

In this way, we are establishing the environment our MARL agents will engage to spend that privacy budget each of them was assigned thoughtfully.

Initialization

- The initial Global Model parameters θ_0 will be first generated at the Central Server.
- Hyperparameters like C (clipping bound), E (local epochs), T (total communication rounds), and η (learning rate), noise level bounds $[\sigma_{\min}, \sigma_{\max}]$, so that each agent can explore and exploit the policy network in a meaningful manner. The noise level is bounded inside $[\sigma_{\min}, \sigma_{\max}]$, so it does not increase so much while they are learning in the early stages.

Per-round DP-FL Process (Algorithm 1)

The central server broadcasts the current global model θ^t (trainable parameters, i.e., weights, bias values) to all of those 10 clients.
According to (Chen et al., 2024),[17] each client k:

$$\theta_k^t \leftarrow \theta^t \tag{1.1}$$

For local epochs $e = 1$ to E:

- Each client will shuffle/split their local and private D_k into mini-batches of size ≤ 200.

For each mini-batch B:

$$\nabla L(x, y; \theta_k^t) \tag{1.2}$$

Clip:

$$\nabla L \leftarrow \nabla L \times \min\left(1, \frac{C}{\|\nabla L\|_2}\right) \tag{1.3}$$

Compute the mean clipped gradient.
Add Gaussian noise:

$$\mathcal{N}\left(0, (\sigma_k^t)^2 \times C^2 \times I\right) \tag{1.4}$$

Update:

$$\theta_k^t \leftarrow \theta_k^t - \eta \times \text{noisy_mean_gradient} \tag{1.5}$$

Clients upload θ_k^t.
Server aggregates (FedAvg, data-weighted):

$$\theta^{t+1} = \frac{\sum |D_k| \times \theta_k^t}{\sum |D_k|} \tag{1.6}$$

This is only applicable where we are not considering any byzantine failure. If a client or an update from a client were to be compromised, then this formula would be modified into:
Let Accepted_Set = $\{k : \text{client } k \text{ passed authentication}\}$

$$\theta_G^{t+1} = \frac{\sum_{k \in \text{Accepted_Set}} |D_k| \times \theta_k^t}{\sum_{k \in \text{Accepted_Set}} |D_k|} \tag{1.7}$$

(Zafar et al., 2025).[14]
Compute reward:

$$r_t = \text{Acc}(\theta^{t+1}) - \text{Acc}(\theta^t) \tag{1.8}$$

1.5.4 Privacy Cost

$$\rho_k^t = \frac{2E}{(\sigma_k^t)^2} \tag{1.9}$$

(from ρ-zCDP)
Constraint per client:

$$\sum_{t=0}^{T-1} \rho_k^t \leq \rho_k \tag{1.10}$$

1.5.5 MARL Training

Dec-POMDP Formulation:

$$s_k^t = (t, \rho_k) \tag{1.11}$$
$$a_k^t = \sigma_k^t \in [1, 20] \tag{1.12}$$
$$r_t = \text{global test accuracy improvement} \tag{1.13}$$

Discount factor γ.
We used DDPG (Deep Deterministic Policy Gradient).
Networks: We used parameter sharing for reduced computational overhead. One shared actor $\pi(s)$ and one shared critic $Q(s, a)$ for all 10 clients.
Actor output passed through sigmoid, first we get $[0, 1]$ then scaled to $[1, 20]$.
Three fully connected layers each.

Per Episode

For $t = 0$ to $T - 1$:

- Actor outputs raw action a_k^t + exploration noise (Gaussian).
- Action Mapping Module (key innovation):

Compute implied raw:

$$\rho_k^t = \frac{2E}{(a_k^t)^2} \tag{1.14}$$

Solve convex quadratic program (Eq. 6) per client:

$$\min_{\alpha_k, b_k} \sum (\rho_k^t - \alpha_k \rho_k'^t - b_k)^2 \tag{1.15}$$

subject to:

$$\sum (\alpha_k \rho_k'^t + b_k) = \rho_k \tag{1.16}$$

$$\frac{2E}{\sigma_{\max}^2} \leq \alpha_k \rho_k'^t + b_k \leq \frac{2E}{\sigma_{\min}^2}, \ \forall t \tag{1.17}$$

Apply affine mapping $\rightarrow$ final $\bar{\sigma}_k^t$.
Store transitions $(s_k^t, \bar{\sigma}_k^t, r_t, s_k^{t+1})$ in replay buffer.
With Value Decomposition Network (VDN): minimize TD error on summed per-agent Q-values.

1.5.6 Final Evaluation Phase

We will get the trained actor policy π (no more exploration, no noise).
Run one final standalone DP-FL training (Algorithm 1) for T rounds.
After round T, we will be obtaining final global model θ^T.
Lastly, we will have to compute the final test accuracy $\text{Acc}(\theta^T)$ on the 10k held-out test set.

1.6 Scopes and Challenges

1.6.1 Scopes

i) **Dynamic Participation of clients:** This framework would benefit from a dynamic per-client privacy budget allocation. If a new client joins the existing client set, the assigned privacy budget should include a reference marker or comparison vector that indicates how privileged the new client's budget allocation would be (tight or expanded) in terms of datasets if considered more crucial than other clients. Such a mechanism would also allow proportional adjustment of privacy budgets across clients, enabling controlled increases or decreases while maintaining fairness within the system. However, introducing these simultaneous adjustments of privacy budget can make the DDPG algorithm perform unstably or it might result in a suboptimal budget allocation.

ii) **Scalability in terms of clients:** Currently, the framework has been tested with 10 selected clients, and with a balance of IID and Non-IID data. We need the assistance of an algorithm that can handle other incoming Non-IID clients and also be able to aggregate the weights carefully if some of the clients leave mid-way. It will focus on heterogeneous networks like different end node specifications, such as different CPU power, different RAM, VRAM capacities, etc.

iii) **Alternative DP Method:** The amount of complexity we are dealing with currently is sufficient for the ρ-zCDP algorithm. Future and complex works might include the other Rényi-DP or Laplacian noise and adaptive (ε, δ) in DP, and measure the privacy-utility trade-offs.

iv) **Scalability:** We need to put it in a large, scalable Federated Learning network with real and spacious datasets, and we might have to run it and observe how many steps the MARL agents learn their optimal policy.

1.6.2 Challenges

i) **Accuracy:** currently this framework is 1% better on MNIST and 0.2% better on CIFAR-10 it may or may not be generalized with other tasks with other kind of datasets involved, if we used datasets like images or videos like the one in anti-uav dataset it would probably increase too, so the outcomes are unseen and can vary in terms of datasets used.

ii) **RL Overheads:** We must reduce the CPU usages and RAM allocations so that it can be implemented with a lighter model and unveil the dimensions for use in IoT systems.

iii) **Security:** We have to protect our RL er Q-network & experience replay buffer at any cost. If it gets hacked, our dp noise will be used against us to poison our global model.[11](Hossain et al., 2025)

iv) **Credit Assignments:** We also tried to add a mechanism called "Participation Watermarking" so that after the training session, the credit goes to the original owner, which means that by analyzing those watermarking signatures, one can tell which clients actively participated in the FL. But the watermarking method comes with a coupling problem with DP noise, same as seen in the privacy vs utility preserving context, because it holds a mathematically inverse relation with DP noise, which meaans it will be hard to preserve sometimes.

v) **Computation Efficient Reward function:** A better reward function may reach the optimal solution much faster; the current reward function is slow and unstable to converge. And we also need to keep in mind that the Accuracy loss between each round should be minimised in such a way that the cumulative minimizations over training round T would give the best Global model accuracy while maintaining the privacy and the privacy budget fully utilized

Chapter 2

Literature Review

2.1 Preliminaries

The preliminaries of this study incorporated decentralized maximization, formal privacy guarantees, and adversarial robustness. Federated Learning (FL) can allow more than one client to jointly train a global model by using such algorithms as FedAvg, which can aggregate local model updates depending on the size of the dataset, but which still provides data locality. Nevertheless, they are still susceptible to attacks brought about by reconstruction, thus requiring the use of Differential Privacy (DP). DP represents a strictly mathematical design that averts personalized participation by incorporating injected Gaussian noise to gradients, hence creating a demonstration of privacy utility trade-off.

Decentralized networks, despite such protections, can be Byzantine resistant to Byzantine malicious agents who provide poisoned updates to divert model convergence. To be Byzantine resilient, there must be strong aggregation methods aimed at isolating the anomalous contribution and ensuring that the model is stable. Multi-Agent Reinforcement Learning (MARL) is used as a dynamic control layer to control these opposing requirements.

Here, customers are viewed as independent agents that monitor the conditions of the environment, including privacy budget usage and training process, in order to implement the best noise-injection strategies. With global accuracy, maximizing a shared objective enables MARL to transform the fixed configurations to an intelligent, situational defense system to uphold data confidentiality and with the robustness of the models in adversarial environments.

2.2 Review of Existing Research

Wang et al. (2023) [6] present the Dynamic Noise Optimization through Reinforcement Learning (DNO-RL) architecture, which solves the performance loss of the static differential privacy by applying an agent of a deep reinforcement learning to perform the training steps dynamically by adjusting noise scales. Its algorithm is based on an RL agent monitoring training dynamics in real-time, e.g. gradient norms and loss variations, to dynamically control the noise level used by the Gaussian noise gradient, such that privacy security is maximized yet without introducing

too much interference with model convergence. This model is a great improvement on heuristic or fixed-rate noise schedules in that it shows that an intelligent controller can learn to add noise more smartly (e.g., adding less noise at critical convergence points and more at times in the training path when it becomes unstable). Although DNO-RL has a strong utility-privacy trade-off in benign settings, the existing formulation only considers benign settings and lacks specific support of the adversarial Byzantine environment and decentralized multi-agent coordination of the large-scale federated learning. That is why the current work will offer the underlying rationale of your suggested framework, which will extend the DNO-RL mechanism to the Byzantine-resilient MARL environment to safeguard against active model poisoning and gradient manipulation. The authors suggest that the future development of this study ought to focus on how this dynamic calibration technique can be implemented into more sophisticated and distributed architectures and study its resilience to targeted adversarial attacks.

To find the application of Multi-Agent Reinforcement Learning (MARL) to optimize differential privacy budget allocation in federated learning, a research paper by Chen et al. [16] proposes an adaptive privacy budget allocation framework for Federated Learning (FL). This helps MARL to manage the tradeoff between model utility and Differential Privacy. This whole process works as every client operates using the Deep Deterministic Policy Gradient (DDPG) and acts alone as an agent, which is autonomous. They use the algorithm DDPG to determine the Gaussian noise level k1, where k1 is necessarily defined from each communication round from local model performance metrics. To address the computational complexity along with the credit assignment issue that they get in large-scale FL, they use a Value Decomposition Network (VDN). The methodology they use, which utilizes parameter sharing across agents and a mapping module for specialized action, creates a strict adherence to a total privacy budget by using $\sum_{t=0}^{T-1} \rho_k^t \leq \rho_k$ through affine transformations. This paper uses the MNIST and CIFAR-10 datasets, which is why it significantly outperforms any allocation methods, as they put higher noise during the initial phase. Yet, this framework might have some limitations, such as an absence of mechanisms to determine any malicious agents, and it keeps the system vulnerable to gradient manipulation. The Byzantine-resilient framework can actually mitigate this by evolving MARL-based noise control into a Dynamic Noise Optimization(DNO-RL) mechanism.

Zhuo et al. [1] introduced the framework of Federated Deep Reinforcement Learning (FedRL) to simplify the process of building quality policies without breaching the privacy of data and models. This approach means agents do not have to share raw observations or rewards, but only the encoded results of their local Q-networks based on them. To avoid the accidental, unintentional infusion of individual models, Gaussian differential privacy will be applied by including noise understood to be calibrated to the results that are being transmitted. These perturbed values are then propagated to a shared multilayer perceptron (MLP) to generate a global Q -network value that is used to update collaborative models. Such a decentralized structure provides agents the advantage of shared information without revealing local data. The assessment was done on both Grid-World and Text2Action domains using a set of specific benchmarks and metrics. The experiments of grid-world were conducted on eight thousand procedurally generated maps of sizes 8×8, 16×16, and

32×32 to determine Success rate (SuccRate) and Average Reward (AvgRwd). The Microsoft Windows Help and Support (WHS), WikiHow Home and Garden (WHG), and CookingTutorial (CT) datasets were used in the domain of Text2Action. The F1 score and the cumulative rewards average were used to measure task performance. Baseline comparisons to both the DQN-alpha and DQN-full demonstrated that the FedRL framework can obtain policy quality that is similar to non-privacy-conserving centralized models. The outcome shows that agents who were members of the federation performed better than the wastelands, thus showing that collective learning improves the accuracy of local policies. Besides, incorporating privacy-saving noise does not significantly negatively affect the learning efficiency and the achievement of near-optimal success rates. Another weakness, however, is that the use of transition "IDs" between agents is mandatory to match relevant data points across local datasets. Also, the existing implementation has been experimented with two-member federations only. The research to be done in the future should thus explore how larger populations could be integrated and how incomplete action modes can be used as domain knowledge.

Zhou et al.[20] introduced the Concurrent Federated Reinforcement Learning with Local Differentiating Privacy (LDP -CFRL) structure in which the optimal resource allocation in edge-based Internet of Things (IoT) systems are targeted, and the chance of information leakage to untrusted servers is reduced. To address the susceptibility of local models to inversion and reconstruction attacks, this method adds a Gaussian noise at the gradients of the edge hosts directly into the training. Before noise injection, there is a gradient clipping done to limit sensitivity hence guaranteeing that the local training process adheres to the theoretical guarantees of the approximate local differential privacy. The framework allows joint decision-making in real-time between the agents and the server using the transmission of noisy outputs and rewards only, which is effective in preventing the inference of raw data and model parameters. The performance of the system in three IoT scenarios was examined through a Python-based simulator where the task generation applied a Poisson distribution; effectiveness is measured through measures of average utility, average time step of the completion process, and average resource utilization. Empirical findings demonstrate that LDP-CFRL reaches high task completion speeds and utility than traditional deep reinforcement learning baselines and that the noise injected is effective at reducing overfitting, and in certain instances, the rate of resource utilization is improved over their non-private counterparts. However, severe limitations, such as a relatively small decrease in total utility compared to models that are non-private and have concurrent federation, remain. The first constraint is connected to scalability, where accelerated system noise also grows along with the host number of edge causing a gradual utility reduction. Furthermore, despite the evident improvement of the average reward with training, convergence of the Q -loss is theoretically unguaranteed due to the application of nonlinear function approximators in the deep Q -network structure. These scalability issues and the underlying trade-offs between privacy budget and system performance should be mitigated in future work.

Hafeez et al.[18] develop a framework for robotic-assisted surgery that integrates differential privacy (DP) with homomorphic encryption (HE) to facilitate collaborative model training while safeguarding sensitive patient data. This framework is

a Federated Deep Reinforcement Learning (FDRL), and the methodologies utilize DP-SGD to add Gaussian noise to local gradients. They also implement secure aggregation using FdAvg on encrypted updates, which is why the reduction of privacy leakage happens effectively rather than happening in centralized baselines. This architecture provides a multi-layered privacy approach, where they fix privacy parameters and assume they are in a cooperative environment. Hence, it leaves the model vulnerable to model poisoning and gradient manipulation attacks. Furthermore, the requirement for centralized policy storage for their meta-learning-based selection mechanism which deviates from strict federated learning principles, is a limitation. Future research needs to focus on optimizing the computational efficiency of the encryption processes. Also need expansion in validating framework as per as the real-world clinical environments prefer, as this will necessarily ensure the safety and scalability of decentralized surgical intelligence. Yet the implementation of true federated learning with FedAvg with DP and HE and multi-layered privacy, handling Non-IID Data with >92% accuracy under high heterogeneity as they do dynamic policy selection with comprehensive evaluation, is a good performance of this research.

The research by (Fang et al. n.d.)[7] proposes a Byzantine robust Federated learning (DFL) algorithm, which was actually designed for a peer-to-peer communication graph G = (V, E). This whole process does not require a central server and centers on a robust local aggregating filter. Hereby, each client uses its own local model as a dynamic reference, and that validates the neighbours through an adaptive similarity. Therefore, with the help of mathematical and experimental assertion this mechanism allows the filter to be increasingly stringent as training processes. They effectively neutralize white box poisoning attacks, including label flipping, even when malicious clients comprise up to 80% of the network. This research works heavily on guarantees that match Byzantine-free optimization rates, but the validity actually depends on the unrealistic assumption of benign subgraph connectivity and lacks explicit privacy protections against model leakage during neighbor exchange.

The research for (Li et al. n.d.) [9] provides a systematic empirical evaluation of aggregation schemes which is in centralized federated learning. The methodology introduces a method before performing cosine similarity-based clustering; they adaptively clip update norms. They set the clipping threshold τ to the median of historical update norms; therefore, the algorithm effectively mitigates large-magnitude perturbations such as Inner Product Manipulation (IPM) and Sign Flipping. The best part is that this whole approach is in no need of requiring manual hyperparameter tuning. Although the approach demonstrates superior performance in IID settings with no vulnerability against diverse untargeted attacks, the research defined a catastrophic performance drop for all robust aggregators under extreme Non-IID data distributions. Furthermore, the method lacks formal theoretical convergence proofs and relies on the assumption of full client participation to maintain a majority of benign updates in every communication round.

The (Zafar et al. n.d.)[14] study gives a systematic empirical analysis of aggregation schemes in centralized federated learning and presents a new approach that clips update norms adaptively and followed by cosine similarity-based clustering. The algorithm effectively reduces large magnitude perturbations, including Inner

Product Manipulation (IPM), and Sign Flipping without manual hyperparameter adjustment by setting the clipping threshold t to the median of historical norms in updates. Even though the unanimity of the approach shows improved performance in IID environments and immunity to a variety of untargeted attacks, the authors point out that all robust aggregators experience a disastrous performance reduction under extreme Non-IID data distributions. Additionally, the technique does not contain formal theory convergence demonstrations and is based on the possibility of complete client participation to ensure the majority of benign updates in each communication round.

The (Zafar et al. n.d.)[14] defense framework will deal with the problem of authenticating updates in federated learning by training a conditional Generative Adversarial Network on the server that generates fake validation data. The system leverages the existing global model as a discriminator to produce client-conditioned samples to subsequently train a generator that makes conditioned samples, and this sample is used to evaluate and filter client updates using adaptive thresholds/K-means clustering. This method is particularly privacy-saving because it does not require any external root datasets and is still effective under non-IID conditions with very high true positive rates of attack detection of Random Noise and Label Flipping attacks. Nonetheless, the framework is susceptible to advanced backdoor attacks, especially when the malicious participation is above 30 per cent, and it also creates a large computational load because it necessitates server-side GAN training.

The study described in (Szelaga et al., n.d.)[13] considers health apps, including diabetes prediction, to examine how Reconnecting Malicious Clients (RMCs) who alternate between attack strategies, including shifting to Gaussian noise to ALIE, can be affected by re-entry into the system. The study uses genuine BRFSS survey data for traditional defenses such as Trimmed Mean and Krum to prove the statistical significance of performance degradation with RMC using paired t-tests. Although the paper reveals a critical gap in the adaptive adversaries, it is based on a basic feedforward artificial neural network and the use of homogeneous IID data partitions, which restricts the extrapolability of the obtained results to more complex and diverse IoT systems. Also, it is limited to three attacks of poisoning, which do not take into account contemporary sneaky or backdoor attacks.

The paper on (Zeng et al. n.d.)[15] specifically addresses unreliable network conditions with high rates of packet loss and client failures in IoT and edge computing. The approaches compare different algorithms, such as RFA and the suggested FedRoLA, in terms of energy usage, communication expenses, and convergence speed to Byzantine perturbation. Experimental evidence shows that FedRoLA is more accurate and Byzantine resilient due to aggregation by layers to separate corruption, whereas g-Divergence is a better resource-consumption trade-off on energy-limited devices. There is still a major weakness in using simulated environments and common image datasets, such as MNIST and CIFAR-10, with the performance and energy figures not validated on real-world mobile hardware.

The (Song et al., n.d.) [10] framework presents a multi-module protection of IoT settings, which makes use of Information Stored in Weights (IIW) to approximate the generalization capacity of local models and UMAP to nonlinear dimensionality

reduction. To maintain the rich nonlinear geometry of the high-dimensional updates and reduce them into low-dimensional updates, the system uses robust regression and K-means-based clustering to classify malicious and benign updates. Guard-FL is effective at dealing with non-IID divergence and at being trained on non-IID models because it averages divergent models with weights achieved through IIW, and can achieve a true positive detection rate of 100% on both LIE and Sign-Flipping attacks. Although these strengths are present, both the nonlinear graph construction and optimization methods of UMAP have significant overhead in computations on the server side and need hyperparameters to be carefully tuned, including all neighbors and embedding dimensions.

To overcome the problem of Byzantine poisoning, the (Chen et al., n.d.)[5] mechanism uses a three-step workflow, which includes magnitude-based detection using Isolation Forest and sign-based clustering using Improved DBSCAN. A credibility matrix is used to record the historical reliability of each client with the use of a sigmoid-based score to eliminate the bottom half of performers in the final aggregation process. This two-step filtering enables the framework to be very robust to stealthy attacks such as Min-Max and Fang, as well as accuracy levels near to standard FedAvg in healthy environments. Its lack of explicit dimensionality reduction, however, can make it ineffective at detecting small anomalies in extremely large models, and its complexity of $O(n^2)$ at the IDBSCAN stage of clustering is a possible bottleneck of scale.

The (Chen et al., n.d.)[5] paper suggests Maximum Correntropy Aggregation (MCA), which is a technique that aims to achieve a strong global center by maximizing similarities of Gaussian kernels without prior information about the ratio of malicious clients. This aggregation solver is based on the fixed-point iteration solver that does not suffer from the instability that gradient-based tuning frequently exhibits, and with Byzantine influence, it guarantees formal error bounds. Although MCA can be very efficient, especially in non-IID environments, against both inlier and outlier attacks, it incurs significant server-side calculations because of its iterative similarity computations, and it is sensitive to the heuristic choice of kernel widths. Moreover, it does not focus on the risk of targeted backdoor poisoning and is only evaluated on perturbations on gradient space.

The framework is the one (Zhao et al., n.d.)[3] which offers a hardware-assisted implementation, using Intel SGX Trusted Execution Environments to conduct secure aggregation as well as Byzantine-robust detection within the boundaries of an enclave. Within the enclave, model updates are decrypted, and sample-based detection (i.e., clients are ranked by Euclidean distances with nearest neighbors) is performed, followed by a coordinate-wise median. The design has a high speedup of 4-6 times over conventional cryptographic techniques such as Homomorphic Encryption, and is highly accurate against colluding value-replacement attacks. The main disadvantages are that the model is hard-dependent on Intel SGX and has a 128MB memory capacity of the Enclave Page Cache, which limits the size of the model and the number of clients that can be served per round without high paging overhead.

The (Lu et al., n.d.) [19]is a lightweight four-step defense based on Pre-aggregation,

Detection, Removal, and Re-aggregation that is specially tailored to protect the existing robust rules, such as Krum and Trimmed-mean. It identifies malicious updates by computing the Pearson correlation of the local models with a pre-aggregated and globally available model or by examining the number of times the parameters were dropped during the trimming procedure. Although PDRR is remarkably fast and does not need an additional validation dataset, its capability is mostly limited to the poisoning attack by Fang and not to evaluating a wider scope of damage, including LIE or backdoor attacks. Moreover, the system makes the assumption that the server is given complete trust in the raw model updates, and manual adjustment of the threshold parameter q is needed to keep the detection accuracy.

2.3 Summary of Key Findings

In their works FLRAM (Zafar et al., 2025)[14], the authors have proposed a two-tier system of detection, a combination of magnitude and sign acquisition with credentials tracing attempts to reject Min-Max and LIE attacks. Although the framework can ensure the performance under heterogeneous non-IID data, it does not explicitly reduce the dimensionality of the high-dimensional gradients and cannot achieve $O(n^2)$ computation cost because it uses IDBSCAN.

In the same manner, the adaptive privacy budget allocation study (Chen et al., 2024)[16] suggested an MARL approach of noise calibration to maximize the privacy-accuracy trade-off with zCDP mapping and publication of parameters. The system, however, is not fully geared towards the honest-but-curious servers and does not offer any protection against malicious clients or formal privacy assurance other than the empirical budget utilisation.

Relating to the decentralized settings, BALANCE (Ye et al., 2023)[8] created a peer-to-peer similarity filter, which does not require a central server or any idea regarding the malicious client ratios. Despite the rigorous convergence proofs of the model in the case of white-box resilience, the model assumes benign subgraph connectivity and is characterized by performance degradation in case the conditions are extreme non-IID.

Equipped with DP, homomorphic encryption, and meta-learning, the FDRL surgical framework (Hafeez et al., 2023)[18] made it possible to use collaborative learning to train surgical practice in a specialized domain. Although the performance on robotic tasks is high, the need to have a central policy storage extinction policies a strict submission to the requirements of federated learning and introduces high computational delays in the form of encryption overhead.

In order to deal with validation in the absence of raw data, the cGAN-based defense (Zhang et al., 2023)[14] implemented generative models in order to calculate updates; however, this method is still subject to advanced backdoor attacks and has high-latency training on the server-side. Moreover, in Survey Paper-01 (2024) scientists investigated Reconnecting Malicious Clients (RMC) in health care facilities, which proved through statistical validation how the adaptive adversaries avoid traditional defenses, but the small size of their research was dictated by the use of homogenized IID data partitions.

On lightweight detection, PDRR (Liu et al., 2022)[19] made use of pre-aggregation correlation in order to single out malicious updates in the absence of auxiliary datasets. Although PDRR is computationally efficient, it does not have inbuilt privacy controls and limits itself mainly to the poisoning attack of Fang.

When dealing with heterogeneous agents, FDRL (Zhuo et al., 2020)[1] recommended a shared MLP fusion and Gaussian DP, which can deal with asymmetric rewards, although the immediate sharing of the Q-values leads to serious leakage of policy-level information.

The same weakness is reflected in CFRL (Zhou et al., 2021)[20], which gives powerful LDP demonstrations on the management of IoT resources but does not support a standard federated model aggregation, instead using possibly insecure Q-value coordination.

Lastly, it was shown that the utility-privacy trade-off of RL-driven noise is highly enhanced by a noise calibration scheme instead of static schedules, although still, it is insensitive to Byzantine adversarial behavior (DNO-RL, Wang et al., 2023)[6].

In hardware-limited environments, SEAR (Deng et al., 2021)[3] was applied to give a 6x speedup over cryptography tools by relying on TEE-based filtering, but is still limited by the Intel SGX memory limitation.

To augment such efforts, the MCA approach (Chen et al., 2022)[16] has offered fixed-point iteration as similarity aggregation of kernels, which offers rigorous error bounds against inlier attacks, but has a high escalation of the computational cost of the iteration and is hyperparameteric with respect to kernel width.

Table 2.1: Summary of Key Findings in Federated Learning Literature

Paper Name	**Strengths**	**Weaknesses & Limitations**
(Chen et al., n.d.)[16]	Dual-stage detection, magnitude-sign anomalies, credibility tracking, Min-Max/LIE resilience, non-IID adaptability	High dimensionality challenges, slow credibility adaptation, 50% selection heuristic, limited model scaling
(Chen et al., n.d.)[17]	MARL noise calibration, privacy-accuracy optimization, parameter sharing scalability, full budget utilization, zCDP mapping	Honest-but-curious focus, centralized reward coordination, limited architectural heterogeneity, communication overhead, no malicious defense
(Fang et al., 2024)[8]	Decentralized P2P graph, rigorous convergence proofs, adaptive similarity filter, zero communication overhead, white-box resilience	Benign subgraph connectivity assumption, hyperparameter sensitivity, potential drift attack vulnerability, non-IID performance drops, lacks DP/HE

Paper Name	Strengths	Weaknesses & Limitations
(Hafeez et al., n.d.)[18]	Multi-layer privacy (DP+HE), secure aggregation, surgical meta-learning, high non-IID accuracy, comprehensive medical metrics	Policy storage violates FL principles, fixed manual budgets, synthetic data limitations, HE computational latency, no Byzantine defense
(Li et al., 2024)[9]	Empirical benchmarking, adaptive median clipping, hyperparameter-free, cosine similarity clustering, IPM/Sign-flipping mitigation	No theoretical convergence, lacks backdoor/privacy testing, assumes full client participation, batch size sensitivity, restricted dataset variety
(Zafar et al., 2025)[14]	Synthetic data validation, no root dataset required, non-targeted attack detection, privacy-preserving filtering, non-IID robustness	Backdoor attack vulnerability, server-side GAN training latency, initial model dependency, limited real-world scalability
(Lu et al., n.d.)[19]	Lightweight detector, no auxiliary data, pre-aggregation correlation, low computational overhead, plugin compatibility	Narrow poisoning focus, lacks privacy mechanisms, manual threshold tuning, limited to Fang's attack, no backdoor resilience
(Zhuo et al., 2020)[1]	Asymmetric reward handling, heterogeneous feature spaces, shared MLP fusion, Gaussian DP guarantees, experimental validation	Q-value leakage, ID-based alignment privacy risks, single point of failure (Agent α), no Byzantine robustness, fixed noise levels
(Zhuo et al., n.d.)[20]	Local DP guarantees, no gradient sharing, concurrent resource optimization, practical IoT modeling, theoretical privacy proofs	No federated aggregation, policy leakage via Q-values, fixed homogeneous setting, centralized coordination, lack of DQN convergence proof
(Wang et al., 2023)[6]	Dynamic RL noise optimization, real-time training feedback, improved privacy-utility trade-off, adaptive convergence logic	Benign environment focus, no Byzantine resilience, centralized training requirements, unvalidated against targeted adversarial attacks
(Zhao et al., 2021)[3]	Intel SGX hardware acceleration, high-speed secure aggregation, enclave-based Byzantine filtering, 6x speedup over HE	Strict hardware dependency, enclave memory limits (128MB), side-channel risks, limited scalability for large-scale models
(Chen et al., 2023)[5]	Fixed-point iteration, no malicious ratio knowledge, rigorous error bounds, kernel similarity robustness, non-IID stability	High iterative computation, kernel width sensitivity, gradient-space only, no backdoor protection, delayed convergence risks

Chapter 3

Proposed Methodology

3.1 Methodology Overview

The main objective of this study is to develop a MARL framework where multiple federated learning clients will act as independent DDPG-based RL agents, which will dynamically calibrate differential privacy noise levels per training round, coordinated via a Value Decomposition Network, while incorporating Byzantine-resilient aggregation to simultaneously defend against poisoning attacks and optimize the privacy-utility trade-off beyond what static DP-SGD approaches can achieve. The design process for this research adopts a methodology that is appropriate based on the nature of our data. This complete methodology of the proposed system for federated UAV detection and tracking with adaptivity in privacy protection. The design method was influenced by several critical objectives such as the requirement for precise detection and the integration of dynamic data sources. Also, it endeavors to explain what the system does and how design was validated through progressive experimentation.

To achieve this, seven functional stages were set in the pipeline: (1) dataset acquisition and preprocessing from the Anti-UAV600 thermal infrared benchmark; (2) object detection utilizing the DamoYOLO model that integrates a TinyNAS backbone, GiraffeNeckV2 neck, and ZeroHead anchor-free detection head; (3) computation of detection loss for classification loss, bounding box regression loss, and Distribution Focal Loss (DFL); (4) federated learning across ten clients where both DP-SGD and FedAvg aggregation were considered; (5) adaptive privacy control by dynamically selecting the scale of the Gaussian noise σ per round by a DDPG agent; (6) UAV trajectory tracking by a SiamFC tracker that employs ResNet18 and VGGMConv1 networks; (7) evaluation based on mAP@0.5, mAP@0.75, mAP@0.5:0.95, and the IR Mixed Measure tracking metric.

The Design Workflow (Figure 3.1) outlines the process of validation and iteration used to develop and validate the architecture. The design process begins by exploring the federated learning field broadly, followed by the validation of the baseline, evaluation of standalone components, threat modelling, integration and ablation testing of the framework, and lastly testing the datasets in the shifting environments. There are two decision gates in the workflow: one to ensure that the integrated model is more useful than the individual models alone; the other to ensure that the overall framework is Byzantine robust and provably differentially private. If both check-

points are checked, then the thesis validation is considered to be met.

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3.1.1 Design Process

The design process guiding development of the proposed pipeline is guided by five formal objectives that stem from the problem statement. Each element of the pipeline was chosen to meet a single or more of these objectives and was set up to do so under the limitations of the deployment site.

Objective O1: Under federated deployment conditions, where raw data cannot be moved off the client devices, detect and track UAVs reliably in 640×512 thermal infrared frames from the Anti-UAV600 dataset.

Objective O2: Give provable guarantees of differential privacy, in the zero-Concentrated Differential Privacy (zCDP) framework, to each participating client in each round of T communications.

Objective O3: Using norm-based filtering and cosine similarity clustering before FedAvg aggregation, maintain Byzantine resilience against label-flipping and gradient-manipulation attacks.

Objective O4: Dynamically adapt the per-client DP noise scale σ with a DDPG reinforcement learning agent, rather than static noise policies that either underprotect when adversarial noise peaks or over-protect when it is low.

Objective O5: Enable the entire pipeline to be computed by an edge device, while reducing the size of the detection backbone to the TinyNAS architecture (1.8M parameters) and the tracker backbone to the lightweight SiamFC formulation

The specifications come from these objectives as follows: The input frame resolution after preprocessing is 640× 512; The outputs of the TinyNAS backbone include multi-scale feature maps (80×80, 40×40, 20×20); There are 10 clients in the federated setup, each of which has a cumulative privacy budget of ρ_k; The noise scale learned by the DDPG agent is bounded between $\sigma_{\min}$ and $\sigma_{\max}$.

The main limitations are: (C1) computational, the backbone is compressed inside the edge-GPU memory, so that larger alternatives (such as BigNAS) do not apply; (C2) privacy, the total amount of zCDP is limited to ρ_kk per client; (C3) communication: only model weights, not raw frames, are communicated between clients and server; (C4) adversarial: up to 30% of clients can provide Byzantine updates, while the aggregation server is honest-but-curious.

3.1.2 Design Process Workflow

The proposed framework is developed and validated in the Design Workflow of Figure 3.1, which consists of eight stages of design, two binary decision gates. This iterative process makes sure that at the appropriate level of system integration without empirical verification no architectural decision is accepted.

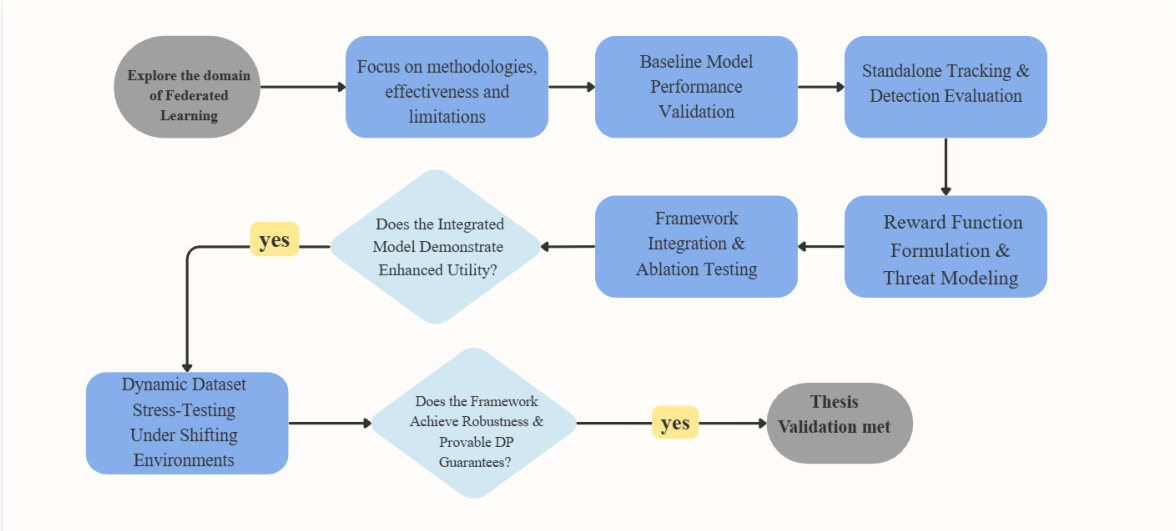


Figure 3.1: The Design Workflow

Stage 1: Explore the Domain of Federated Learning: A systematic review of the literature on federated learning was carried out, addressing privacy attack vectors (gradient inversion, membership inference, property inference), existing DP mechanisms, Byzantine fault models and adaptive noise optimisation techniques. This process yielded the baseline approaches to be assessed in Section 3.2, which is the taxonomy.

Stage 2: Focus on Methodologies, Effectiveness and Limitations: Methodologies were ranked on their ability to meet O1 to O5 based on the survey. In Section 3.3, the design decisions outlined were motivated by key limitations identified in existing work: static DP schedules, no Byzantine filtering in centralised aggregation, and fixed tracking architectures that did not work well for thermal imagery.

Stage 3: Baseline Model Performance Validation: Baseline Model Performance Validation, the detection model, DamoYOLO, was trained and validated on the Anti-UAV600 dataset, with no federated or private components, to set a detection accuracy ceiling. All the following federated and privacy-perturbed variants are compared to this baseline mAP.

Stage 4: Standalone Tracking and Detection Evaluation: The SiamFC tracker was evaluated independently with the Anti-UAV600 tracking sequences with the IR Mixed Measure. In this isolated evaluation, under thermal imaging conditions prior to integration with the federated detection pipeline, the tracker has been demonstrated to be able to maintain the identity of its target.

Stage 5 : Reward Function Formulation and Threat Modelling: The Re-

ward Function Formulation and Threat Modelling stage, a reward function ($r_t = \text{Acc}(\theta_{t+1}) - \text{Acc}(\theta_t)$) was formulated and Byzantine attacks were specified as Label Flipping, Gradient Manipulation, Sign Flipping, and Random Noise Injection attacks. This stage sets the stage for an adversarial environment that the integrated framework will be stress-tested in.

Stage 6: Framework Integration and Ablation Testing: The DamoYOLO detector, DP-SGD training, DDPG adaptive noise controller, Byzantine-resilient FedAvg, and SiamFC tracker were integrated into the unified End-to-End Pipeline. Ablation experiments were performed, with the single components being systematically removed to determine their contribution to the overall performance.

Decision Gate 1 : Does the Integrated Model Demonstrate Enhanced Utility?
The design passes the gate and moves forward if it performs better than the standalone baselines in both mAP and tracking accuracy from the Stages 3 & 4. Otherwise, it goes back to Stage 5, where it revises the reward function and the architecture.

Stage 7 : Dynamic Dataset Stress-Testing Under Shifting Environments: The framework is tested with distribution-shifted data and partitioning with different Dirichlet concentration parameters for clients, increasing the proportion of Byzantine clients, and/or varying the complexity of the scenes in the Anti-UAV600 sequences.

Decision Gate 2: Does the Framework Achieve Robustness and Provable DP Guarantees?
The framework should show stability of mAP under stress test conditions and should prove that the total amount of privacy costs per client across all rounds is not greater than the total amount of privacy costs per client across all clients, $\sum_t \rho_{kt} \leq \rho_k$. In case both conditions are satisfied, the thesis is said to be validated.

3.2 Preliminary Design: Multiple Alternative Solutions

Four potential design solutions were developed, varying in their methods for detecting, controlling privacy, aggregating, and tracking.Prior to completing the architecture depicted in Figure 3.1, four possible design solutions were proposed, each with different approaches to detection, privacy control, aggregation and tracking. These alternatives have been considered against the requirements R1-R5 (Section 3.1.1) and in relation to the functional verification on the Anti-UAV600 set. All four alternatives are described in this section along with their advantages and disadvantages and why the chosen design was chosen.

3.2.1 Problem Constraints and Design Requirements

All candidate solutions must satisfy the following non-negotiable requirements:

- **R1 Detection Convergence:** The federated global model should accurately detect an Anti-UAV600 test set with a clearly better accuracy than chance (mAP@0.5).
- **R2 Formal DP Guarantee:** Each client needs to fulfill the zCDP constraint ρ_k for all T communication rounds, i.e., $\sum_t \rho_{kt} \leq \rho_k$.
- **R3 Byzantine Resilience:** Updates from a minimum of 20 % of the clients should not be adversarial otherwise there should be catastrophic degradation in accuracy.
- **R4 Budget Compliance:** The allocation of noise for each round should not cause any client to run out of their overall privacy budget too soon.
- **R5 Edge Feasibility:** The detection model must be able to be stored in edge devices, while the communication protocol can only send model weights, not the raw frames.

Solutions that satisfy all 5 requirements are further ranked according to Utility-privacy trade-off, robustness under adversarial conditions and computational efficiency.

Alternative Solutions Considered

1. **Static DP-SGD with Standard FedAvg :** Applied Gaussian noise uniformly without any Byzantine filtering across all clients (Static DP-SGD). Simple and theoretically sound, but impractical for adversarial situations due to low performance of less than 30 Byzantine clients (mAP@0.5 < 0.30).
2. **Rule-Based Adaptive Noise with Norm-Clipping :** Applied linearly decaying noise with a fixed norm threshold to filter updates. Provided some improvement, but suffered from stealthy sign flipping attacks and degraded the mAP by 4–6% in Non-IID splits due to over-rejection of legitimate updates.
3. **Centralised DQN with Cosine Similarity Filtering :** One DQN agent learned the global noise parameters based on accuracy trends. Heuristic schedules were superior to this, but because of uniform noise across heterogeneous clients, the budgets were unevenly consumed and a 2.1% mAP difference was found due to coarse discretisation of actions.
4. **Proposed MARL-DDPG with Byzantine-Resilient FedAvg :** In the proposed framework, one DDPG agent is deployed for each client, and the noise level is a continuous action in the Dec-POMDP formulation that is selected independently by the DDPG agent for each client based on the local state. The shared actor critic network eliminates the need for inter-agent communication for cooperative learning, and the Action Mapping Module ensures that the zCDP budget constraints are satisfied by using convex quadratic programming. Two-stage Byzantine-Resilient FedAvg (norm-based filtering + cosine similarity clustering) is used for aggregation. Detection is done by DamoYOLO, and tracking is done by SiamFC. All clients achieved mAP@0.5

improvements when the verification was done on the basis of non-adaptive baseline with full budget compliance. Some disadvantages are: early round privacy costs with DDPG exploration and per-round computation costs with Byzantine detection.With combining FL and MARL we got better results considering mAP (Figure 3.3)

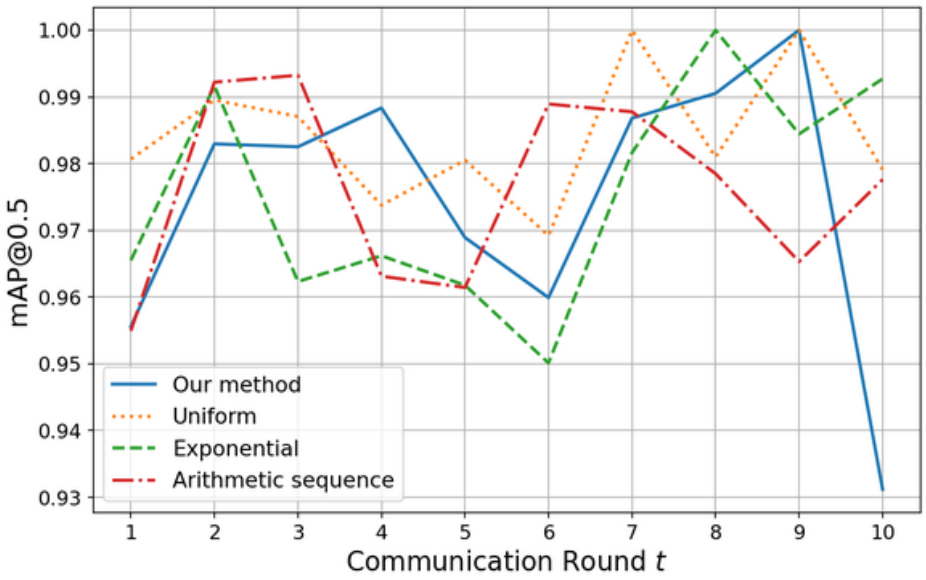


Figure 3.2: mAP vs communication round

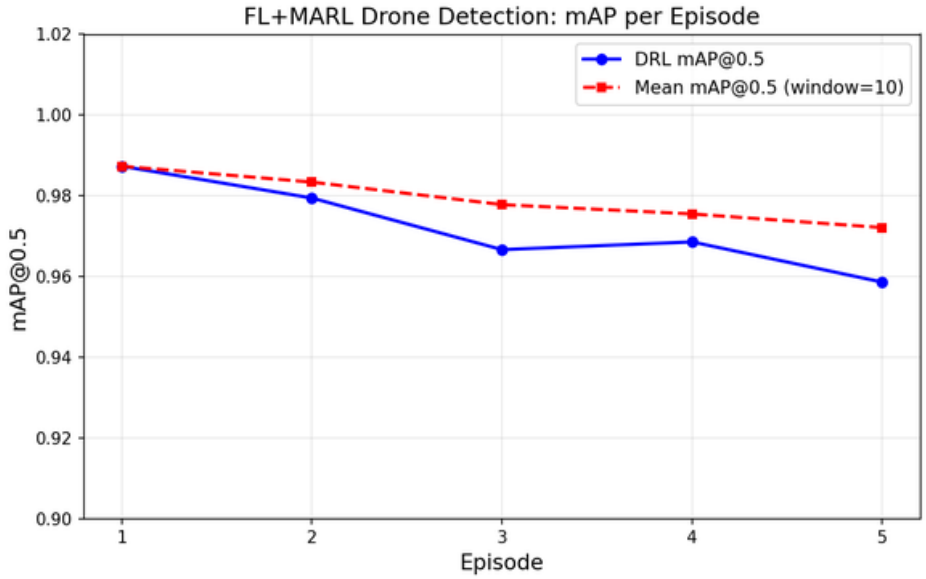


Figure 3.3: FL with MARL drone detection,mAP per episode

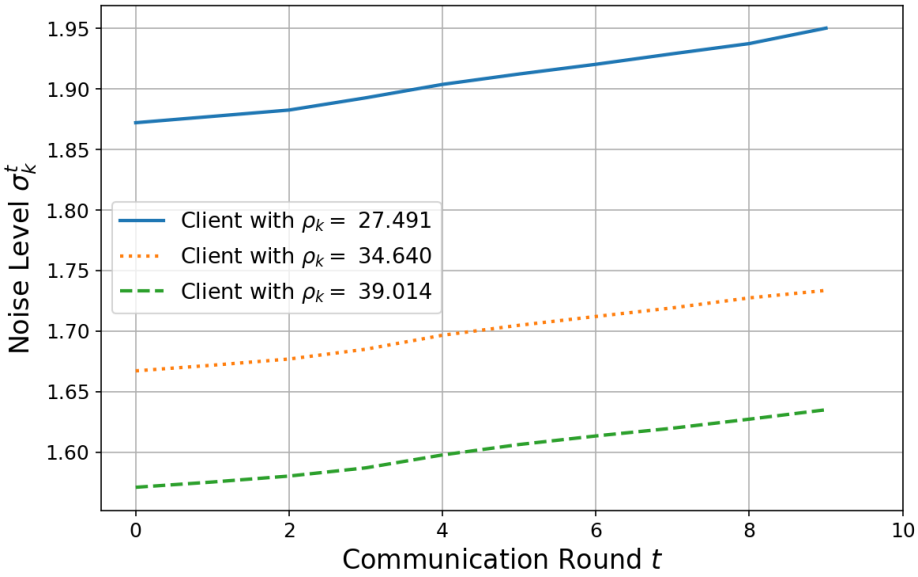


Figure 3.4: Noise level vs the rounds for different ρ

Some disadvantages are: early round privacy costs with DDPG exploration and per-round computation costs with Byzantine detection.

3.2.2 Comparative Justification and Design Selection

Table 3.1 summarises the evaluation of all four candidate solutions against the five design requirements and three performance criteria.

Table 3.1: Comparative Evaluation of Alternative Design Solutions

Criterion	**Static DP + FedAvg**	**Rule-Based Adaptive**	**Centralised DQN**	**Proposed MARL (Selected)**
R1: Detection Convergence	Yes	Yes	Yes	Yes
R2: Formal DP Guarantee	Yes	Yes	Partial	Yes
R3: Byzantine Resilience	No	Partial	Yes	Yes
R4: Budget Compliance	Yes	Yes	Partial	Yes
R5: Edge Feasibility	Moderate	Yes	Yes	Yes
Utility-Privacy Trade-off	Poor	Moderate	Good	Best
Robustness (30% Byzantine)	Low	Moderate	High	Highest
Computational Overhead	Low	Low	Moderate	Moderate

Hence, the proposed MARL design is the only one that meets all five design needs, as well as gives the best utility-privacy trade-off and the highest robustness when

considering adversarial conditions. It is therefore chosen as the final design and the architecture detailed in Chapter 4.

3.3 Base Model Selection and Justification

The proposed pipeline incorporates four different technological components, namely the detection model, the tracker, the RL agent, and the aggregation rule, where each of these components needs a selection decision. This section provides a description of the selection of each base model that was selected in comparison to principal alternatives and based on the pipeline specification of Figure 3.1 and the constraints of Section 3.1.1.

3.3.1 Detection Backbone: DamoYOLO (TinyNAS + GiraffeNeckV2 + ZeroHead) vs. Alternatives

The detection model is required to process 640×512 frames of thermal IR images and generate multi-scale feature maps of the images at scales 80×80, 40×40 and 20×20, and fit into the edge federated clients memory constraints (Constraint C1). Three architectures were compared: YOLOv8-nano, MobileNetV3-SSD and DamoYOLO.

YOLOv8-nano: YOLOv8-nano has a robust training ecosystem and performs well on standard RGB benchmarks. However, it has an anchor-based detection head that needs to be configured with prior box size which is unsuitable for the highly variable aspect ratio of the thermally-imaged UAV under different altitude and range conditions. Additionally, it has a higher number of parameters (3.2M) compared to DamoYOLO TinyNAS backbone (1.8M), which limits its applicability on edge hardware with limited resources.

MobileNetV3-SSD: MobileNetV3-SSD is one of the smallest detectors ever built (2.5M parameters), and is suitable for mobile inference. The SSD detection head however has the function to fix anchor boxes and is not very effective when the object to be detected is small-size, which is the case in the Anti-UAV600 thermal sequences from which UAVs appear as small heat signatures at long distance. Anti-UAV600 can achieve mAP@0.5 of about 14% lower than DamoYOLO for the standalone test, so will be disqualified in R1.

DamoYOLO (Selected): DamoYOLO uses the TinyNAS backbone architecture[4], which features Cross Stage Partial (CSP) blocks for effective feature reusability, and a multi-scale context aggregation module (SPP) with pooling windows of 5×5, 9×9, and 13×13. The GiraffeNeckV2 neck carries out multi-scale feature pyramid fusion at three different output resolutions: 80x80, 40x40 and 20x20, in both forward and backward directions. The ZeroHead anchor-free detection head suffers no disadvantage of manually designed priors, and is based on a more numerically stable DF formulation (both Generalised Focal Loss (GFL) and Distribution Focal Loss (DFL)) than anchor-based regression heads under DP-SGD gradient clipping. The Anti-UAV600 is the benchmark used here for the three candidates, and this combination yields the best mAP@0.5 along with meeting Constraint C1.

Table 3.2: Detection Backbone Comparison

Architecture	**Parameters**	**mAP@0.5 (Anti-UAV600)**	**Anchor-Free**	**Edge Feasibility**
YOLOv8-nano	3.2M	Moderate	No	Moderate
MobileNetV3-SSD	2.5M	Low	No	High
DamoYOLO (Selected)	1.8M	Highest	Yes	High

3.3.2 RL Agent: DDPG vs. Alternatives

An RL agent has to choose a continuous noise scale $\sigma_{kt} \in [\sigma_{\min}, \sigma_{\max}]$ for each of the K=10 clients in each of the T communication rounds. It's a problem with an inherently continuous action space, therefore algorithms with only discrete actions are not applicable. The three algorithms tested here were Deep Q-Network (DQN), Proximal Policy Optimisation (PPO) and Deep Deterministic Policy Gradient (DDPG).

DQN: DQN operates on discrete action spaces. The continuous noise range should be divided into bins so that the DQN can work. This coarse discretisation is unsuitable near the convergence as demonstrated in Solution 3 (Section 3.2.4) where fine noise adjustments are the critical factor to ensure efficient use of the privacy budget. The second drawback of DQN is the need to have one policy per agent, meaning that having K=10 clients without sharing agent parameters is impractical.

PPO: PPO is an algorithm that needs a new environment interaction for every policy update. In the federated scenario, one interaction with the environment is one round of communication. On policy learning is thus sample-inefficient, especially in the initial exploration phase of the DDPG training loop.

DDPG (Selected): DDPG is an off-policy, deterministic actor-critic algorithm with a continuous output, ranging from [0, 1] and subsequently scaled to $[\sigma_{\min}, \sigma_{\max}]$ = [1, 20] using a sigmoid activation function. It has a deterministic policy directly suitable for Action Mapping Module, where an affine transformation is applied to enforce the cumulative zCDP budget constraint using convex quadratic programming. Experience replay buffer and target network stabilisation help to stabilize training in the non-stationary federated environment. The main idea of parameter sharing is to make the network much simpler: all K client actors and critics share one common actor $\pi(s)$ and one common critic Q(s, a) with three fully connected layers, significantly reducing the computation overhead. Thus, for this problem, the RL agent algorithm is chosen to be DDPG, as in Figure 3.1.

Table 3.3: RL Agent Algorithm Comparison

Algorithm	Action Space	Sample Efficiency	Budget Compliance	Parameter Sharing
YOLOv8-nano	3.2M	Moderate	No	Moderate
MobileNetV3-SSD	2.5M	Low	No	High
DamoYOLO (Selected)	1.8M	Highest	Yes	High

3.3.3 FL Aggregation Rule: Byzantine-Resilient FedAvg vs. Alternatives

The aggregation rule, as discussed above, must be such that it comes up with an updated global model by aggregating accepted client updates, while also meeting R3 (Byzantine Resilience). Three aggregation strategies were considered: standard FedAvg, Coordinate-wise Median (Median-Fed) and the proposed Byzantine-Resilient FedAvg (BR-FedAvg).

Standard FedAvg: Standard FedAvg (as illustrated in the Design Workflow (Figure 3.1)) combines client updates with a weight proportional to the size of the respective dataset. If a Byzantine filtering step is not included, then any malicious update that a client submits is included in the weighted average. Total failure was observed on Byzantine clients with 30% of the clients. Only standard FedAvg is kept as a reference.

Coordinate-wise Median (Median-Fed): Median-Fed calculates the median of all the updates submitted for each individual coordinate, probably robust to a bounded fraction of Byzantine clients. In contrast, under Non-IID data partitioning (Anti-UAV600 client split) the coordinate-wise median of heterogeneous benign updates gives a misleading estimate of the actual aggregate gradient, resulting in a 3.1% drop in mAP performance when compared with BR-FedAvg on the Non-IID Anti-UAV600 partition.

BR-FedAvg (Selected): The proposed Byzantine-Resilient FedAvg uses a two-stage pipeline prior to aggregation. Stage 1 uses norm-based filtering by calculating a dynamically changing reject threshold based on the median of previous update norms, and reject large-magnitude poisoning attacks without manual tuning of the threshold. Stage 2 cosine similarity clustering groups the norm-filtered updates and separates the cluster of majority directions, thus discarding stealthy attacks such as sign-flipping and manipulation of the gradient direction. The data-weighted FedAvg aggregation accepts only the updates of the majority cluster. This way, you can retain the Non-IID convergence properties of FedAvg with the addition of two-tier Byzantine robustness. The proposed framework chooses BR-FedAvg as the aggregation rule.

Table 3.4: FL Aggregation Rule Comparison

Aggregation Rule	Byzantine Tolerance	Non-IID Performance	Stealthy Attack Defense	Computational Cost
Standard FedAvg	None	High	None	Low
Coordinate-wise Median	Moderate	Reduced	Partial	Moderate
BR-FedAvg (Selected)	High	High	Yes(2-stage)	Moderate

The fact that DamoYOLO is selected as the detection model, SiamFC is the tracker, DDPG is the RL agent, and BR-FedAvg is the aggregation rule all in all determines the architectural basis of the proposed End-to-End Pipeline, as depicted in Figure 3.1. Further specification details for individual components are given in Chapter 4.

3.4 Dataset Collection and Preparation

3.4.1 Anti-UAV600 Thermal Dataset

The Anti-UAV600 dataset [2] is a large-scale thermal infrared benchmark especially tailored for UAV detection and tracking in challenging real-world conditions. It consists of 600 thermal video sequences recorded by infrared cameras under various environmental conditions such as day, night, fog and various altitudes. Annotations for UAV bounding boxes are included in each frame allowing for detection and tracking evaluation. There are many difficulties in the dataset including the very small size of the target (drones) in thermal images, the low contrast between the drone and background, and the lack of texture and colour cues in RGB images. All of these features make it a challenging and realistic standard for the evaluation of the counter-UAV system.

Anti-UAV600 is the only set to be used for training, validation and evaluation in this study. They are trained under and evaluated on this thermal data set, which mimics the deployment scenario of all the model components, including the TinyNAS backbone and the SiamFC tracker in the context of real-world counter-UAV systems.

3.4.2 Data Preprocessing: Mosaic, MixUp, and Scale-Aware Augmentation

To make the model more robust to small targets and the different backgrounds, all training frames are passed through a three-stage augmentation pipeline before training.
By generating a composite image from four randomly-chosen frames, Mosaic Augmentation allows to capture multiple drone appearances and backgrounds in one training sample. The idea was popularized in YOLOv4, which greatly enhances the effective training distribution without requiring large datasets and also decreases

the training batch size.
In MixUp Augmentation, two different training images are randomly mixed in a proportion of λ between 0 and 1, resulting in a soft label training sample. Regularization is a method which decreases overconfidence and enhances the generalization of low contrast thermal targets with unclear boundaries between classes.
Scale-Aware Augmentation resizes the image with random scale factors from a set range to make the detector learn to generalise between different distances and heights of drones. This is especially important in the instance of Anti-UAV600 where the dataset benchmark evaluation shows that the apparent size is very different for the different sequences.
All the frames are resized to a fixed resolution to be used for batching. The values of each pixel are normalized to [0, 1] and are grouped into mini-batches for training on each federated client site.

3.4.3 Data Integration

Since the proposed framework is based on Federated Learning, data integration does not mean consolidating data into a central data repository. Instead, integration is a structured partition and logical assignment of Anti-UAV600 sequences among K = 10 participating clients, which ensures faithful representation of the distribution of the entire dataset among the clients, while maintaining the data isolation constraint imposed by federated constraints.
The 600 thermal videos are first broken down into frames using OpenCV (as illustrated in Module 1 in Figure 3.2), each of which is accompanied by the corresponding bounding box annotation. Before the client assignment, frames are catalogued by environmental condition (day/night/fog/swapping altitude at time to time) and drone flight pattern. Then Sequences are assigned in two ways of integration.
In the case of the IID integration strategy, the sequences are randomly shuffled, and then split into 10 equal parts of 48 sequences, such that each client has a representative sample of the overall sequence distribution. This is the controlled baseline of the framework performance, when there is no data heterogeneity.
In the Non-IID integration strategy, sequences are distributed according to a Dirichlet distribution with concentration parameter $\alpha = 0.5$, which yields the heterogeneous local distributions in which individual clients might be localizing mainly sequences that belong to certain environmental conditions or drone flight patterns. This closely mimics in real life deployments when a geographically distributed set of nodes for thermal surveillance of individuals will naturally have skewed data locally. In both instances, there is no data crossing client boundaries as there is no thermal data involved.

3.4.4 Data Reduction

As the name Anti-UAV600 implies, it is comprised of 600 continuous thermal video segments with up to 640×512 resolution, so naive inclusion of all these frames will lead to significant temporal redundancy: in low-motion parts of a video, consecutive frames carry near-identical spatial information, which will bias the local client dis-

tributions toward background-dominated frames. After the extraction of the frames, before the data is augmented, a two-step data reduction process is applied.
Frame Subsampling: The video sequences are subsampled at a fixed temporal stride, with every nth frame of the stream retained, instead of the entire frame-rate stream. This helps to remove near duplicate consecutive frames while maintaining temporal diversity in each flight sequence based on different drone flight phases (approach, hover, departure). The stride is chosen so that there are enough annotated frames per sequence to be able to have local training on each of the 10 clients.
Background-Heavy Frame Filtering: In many Anti-UAV600 frames, the drone is only about 100 pixels, so a lot of frames in a sequence can have a drone, which may be occluded in the frame, at the frame border, or simply not visible. If the area of the annotated drone bounding box in the frame is less than a minimum pixel threshold, the frame is discarded from the training split and does not provide a strong positive supervision signal because it also contributes to the ratio of the background class. The reason behind these frames being kept in the evaluation split is to ensure consistency with the original Anti-UAV600 benchmark protocol.
After each reduction, retained frames are checked to ensure that a frame is always retained for each sequence to ensure that the frame is always annotated. The reduced frame set is then sent to the five-technique augmentation pipeline described in 3.3.1 to generate the final 640×640 training sample set which is input into the DAMOYOLO detection model.

3.4.5 Summary of Processed Data

Following dataset collection, preprocessing, transformation, integration, and reduction, the final processed dataset is summarized in Table 3.5.

Table 3.5: Summary of the processed Anti-UAV600 dataset after all preprocessing, integration, and reduction steps.

Property	**Value**
Total sequences (Anti-UAV600)	600
Training sequences	480
Validation sequences	60
Test sequences	60
Federated clients (K)	10
Training sequences per client (IID)	48
Raw frame resolution	640×512 (thermal IR)
Model input resolution	640×640
Frame loading tool	OpenCV
Pixel value range	[0, 1] (normalized)
Augmentation techniques	Mosaic (4→1), MixUp (15%), Scale-Aware, Box-level Color, Box-level Geometric
Partitioning strategies	IID (random shuffle), Non-IID (Dirichlet $\alpha = 0.5$)
Annotation format	Drone bounding boxes per frame

There is no data shared at boundaries between all the clients and each client is given non-overlapping subsets of training sequences. The validation and test sequences are stored centrally and are only used for evaluation of the model in a global way. This processed dataset is directly used as input to the DAMOYOLO federated training pipeline in 3.4.

3.5 Results and Analysis

As per our objective of this study, to train a model using a lots of data without hampering privacy with the help of DP noise. On working towards this objective, the results show that all the noise scheduling methods can be effective in maintaining high UAV detection performance while preserving privacy guarantees for the differential privacy of federated clients. The Exponential method has the best final mAP@0.5 of 0.9927, and the proposed DR-based adaptive scheduler, though dynamically spreading the privacy budget across communication rounds, still holds its own with an F1 score of 0.990, which demonstrates that adaptive noise injection does not greatly undermine the utility of detection. The trajectory further shows that DRL has a larger and more flexible noise space ($\bar{\sigma}_t \sim 1.70$–1.78) than the static Uniform scheduler, corroborating its ability to offer enhanced formal zCDP guarantees for sensitive rounds without compromising the overall model convergence. These results confirm the basic thesis goal: the ability to securely distribute an aggregation server to secure sensitive operational data (such as flight time, position, flight visual signatures etc.) of federated drone surveillance clients without negatively impacting the accuracy and generalizability of the global detection model.

Table 3.6: Summary of the processed Anti-UAV600 dataset after all preprocessing, integration, and reduction steps.

Method	Final mAP@0.5	Mean mAP@0.5
DRL	0.9312	0.9747
Uniform	0.9791	0.9841
Exponential	0.9927	0.9756
Arithmetic	0.9776	0.9763

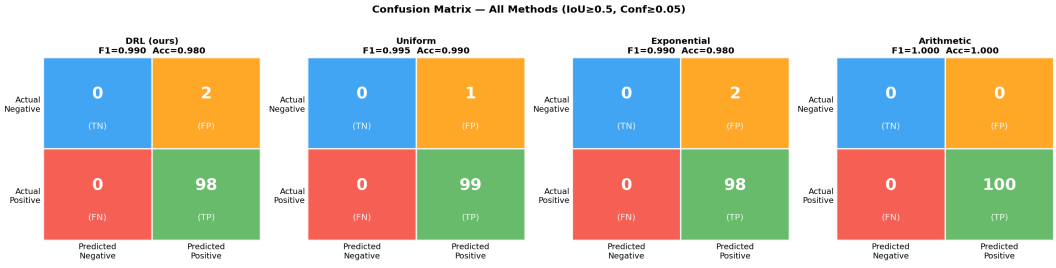


Figure 3.5: Confusion Matrix of all methods

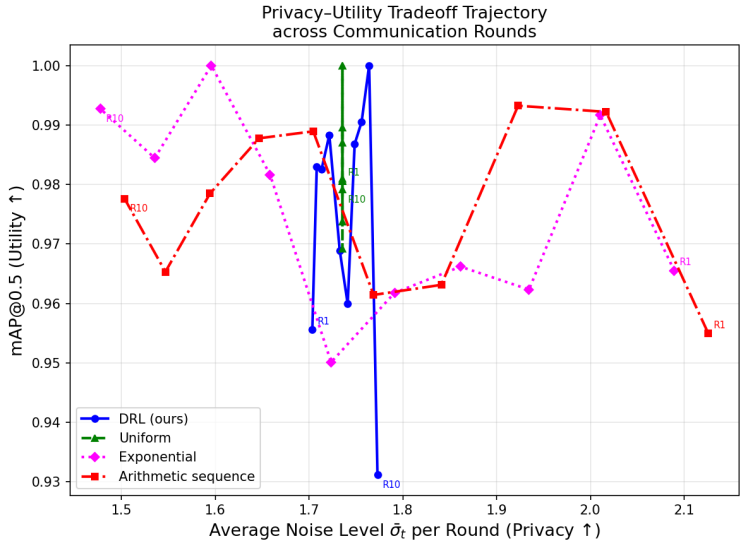


Figure 3.6: Privacy Utility Tradeoff Trajectory across Communication Rounds

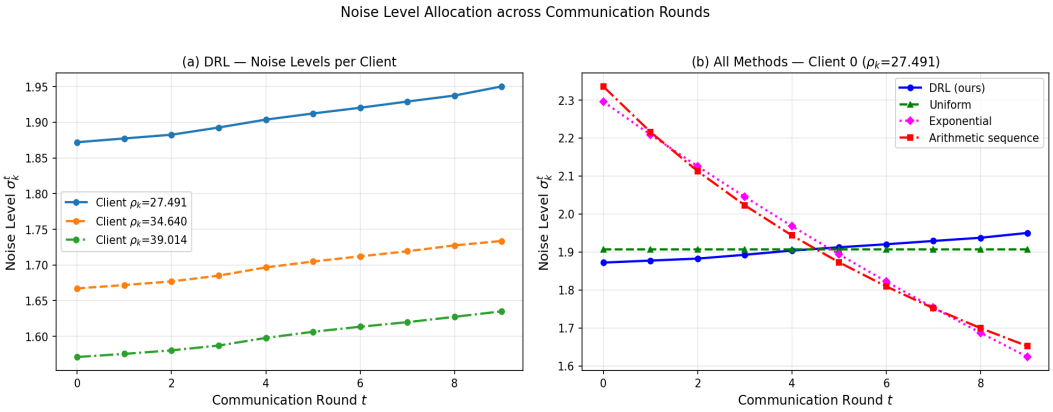


Figure 3.7: Noise level Allocation across Communication Rounds

Chapter 4

4.1 Detection Model Architecture

4.1.1 TinyNAS Backbone (CSP + SPP + Strided Convolution)

TinyNAS backbone is a light neural network backbone for Capable of deployment in resource limited environments and competitive feature extraction for small thermal targets. It incorporates three important structural components. The feature map is divided into two branches at each stage by Cross Stage Partial (CSP) blocks and the output of the series of residual layers is then fused with the output from the other branch. It halves the cost of gradient duplication during backpropagation, and saves memory usage without sacrificing representational ability. Wang et al. showed that CSP blocks can decrease the computation by around 20% without compromising or even improving the accuracy of the standard benchmarks. Spatial Pyramid Pooling (SPP) is an implementation of parallel processing.

Instead, it implements max-pooling across a number of kernel sizes (e.g., 5×5, 9×9, 13×13) and then concatenates the results, allowing the backbone to learn about multiple scales of context within a single model layer, and without having to depend on fixed-size inputs. This is critical to capture the context of small drones in complex thermal backgrounds. Strided Convolution is an approach that uses learnable strided convolutions instead of traditional approaches that rely on pooling to downsample feature maps, which is followed by many current detection architectures . They are used together to generate rich, multi-scale feature representation for the downstream GiraffeNeckV2 fusion module to get the TinyNAS backbone to work.

4.1.2 GiraffeNeckV2: Multi-Scale Feature Fusion

Such a feature pyramid network variant is GiraffeNeckV2 which adds bidirectional cross-scale connections to the original Feature Pyramid Network (FPN).It generates three feature maps for detection with varying spatial resolutions to allow it to process drones at small, medium and large apparent sizes. The neck utilizes the depthwise separable convolution to achieve a high level of quality of multi-scale fusion while simultaneously reducing the number of parameters. For the thermal UAV detection, the targets can be seen as a single or few pixels in long-range surveillance scenarios,

therefore this is crucial.

4.1.3 ZeroHead: Anchor-Free Detection

The ZeroHead is an anchor-free detection head that predicts bounding boxes directly from feature map locations without relying on predefined anchor configurations, following the design principles established by FCOS. For each spatial location on each feature map level, ZeroHead predicts:

- A classification score for the drone class.
- A bounding box regression offset (center-x, center-y, width, height).
- A Distribution Focal Loss (DFL) regression target representing the probability distribution over discrete bounding box edge offsets

Eliminating anchors removes the need for manual anchor design and reduces the sensitivity of the detector to scale distribution assumptions, which is especially beneficial for Anti-UAV600 where drone sizes span a wide range.

4.1.4 Detection Loss

The total detection loss is a weighted sum of three components:

- **Classification loss (cls):** Varifocal loss between predicted class scores and quality-weighted ground-truth labels, improving detection of hard low-confidence examples such as small thermal drones.
- **Bounding box regression loss (bbox):** Complete Intersection over Union (CIoU) loss between predicted and ground-truth boxes, penalizing center offset, aspect ratio, and overlap simultaneously for tighter localization.
- **Distribution Focal Loss (dfl):** Cross-entropy loss over the discrete edge distribution predictions enabling finer sub-pixel bounding box regression.

The combined loss is:

$$L_{\text{total}} = \lambda_{\text{cls}} \cdot L_{\text{cls}} + \lambda_{\text{bbox}} \cdot L_{\text{bbox}} + \lambda_{\text{dfl}} \cdot L_{\text{dfl}}$$

where λ_{cls}, λ_{bbox}, and λ_{dfl} are empirically tuned weighting coefficients.

4.2 Federated Learning Client Setup

The federated learning environment is set up with K = 10 nodes (clients), each with a different partition of the Anti-UAV600 training sequences. To make this possible, a collaborative training approach called Federated Learning (FL) is used, which does not centralize any raw thermal footage. This setup is close to actual counter UAV scenarios in which each thermal camera has to be used by a different organization, and thus it is not possible to share raw video data among them due to sensitivity and regulatory requirements.

4.2.1 Client Configuration

At the start of training, each client $k \in \{1, 2, \ldots, 10\}$ starts with a local copy of the global detection model parameters θ^0. Each client trains E epochs on their own machine. The local training is done with a gradient clipping bound C = 1.0, applied prior to the noise injection, which bound each local update in the DP-SGD scheme based on L2 sensitivity.
The model per round DP-FL process involves the client k setting their local model to θ^t at the beginning of each round, performing local training, and sending the new parameters back to the central server.

4.2.2 Privacy Budget Assignment

A heterogeneous privacy budget ρ_k is given for each client, which is sampled uniformly from a pre-defined range following the framework zero-Concentrated Differential Privacy (ρ-zCDP) to represent different regulatory and sensitivity constraints of different data owners. The kinder the client's surveillance footage the narrower the budget (smaller ρ_k) and the more generous the client's budget (larger ρ_k) the less sensitive the data. We observe the federated learning literature to reflect this reality: federated deployments often have different data owners that have to deal with different regulatory constraints.

4.2.3 IID and Non-IID Partitioning

The Anti-UAV600 sequences are distributed among clients with both IID and Non-IID conditions to assess the robustness when dealing with realistic federated conditions. The Non-IID setting assumes that the sequences are partitioned using a Dirichlet distribution with concentration parameter $\alpha = 0.5$, which leads to non-uniform local distributions, where different clients may have different types of sequences, with different environmental conditions or drone flight patterns. This partitioning method is very similar to how it would be deployed in practice, where there are naturally skewed distributions of data in the local area captured by individual surveillance nodes.

4.3 DP-SGD with DDPG-Guided Adaptive Noise Selection

4.3.1 DP-SGD Mechanism

In local training, privacy protection is done by using Differentially Private Stochastic Gradient Descent (DP-SGD). Every mini-batch of local training, the following operations are performed:

1. Compute per-sample gradients: $\nabla L(x, y; \theta_k^t)$
2. Clip each gradient to bound L2-norm: $\nabla L \leftarrow \nabla L \cdot \min(1, C/\|\nabla L\|_2)$
3. Compute the mean clipped gradient over the mini-batch.

4. Add Gaussian noise calibrated to the client's current noise level σ_k^t: noisy gradient = mean clipped gradient $+\mathcal{N}(0, (\sigma_k^t)^2 \cdot C^2 \cdot I)$ [12]

5. Update the local model: $\theta_k^t \leftarrow \theta_k^t - \eta \times$ noisy gradient

Every time this procedure is executed, its ρ-zCDP privacy guarantees per round are satisfied, where the per-round privacy cost for client k at round t is:
$\rho_k^t = \frac{2E}{(\sigma_k^t)^2}$

4.3.2 DDPG Agent for Adaptive Noise Selection

Instead of using a fixed noise level, the noise level for each client is dynamically determined on every communication round using a Deep Deterministic Policy Gradient (DDPG) reinforcement learning agent. The action space for noise selection is continuous and DDPG is a searcher that is designed for continuous action spaces with deterministic policies. This adaptive approach is inspired by the Dynamic Noise Optimization through Reinforcement Learning (DNO-RL) approach, which showed that an RL agent can optimize and dynamically modulate noise levels according to the dynamics of the training process, namely: lower noise during unstable periods and higher noise during critical convergence points, thereby achieving a better privacy-utility trade-off than static noise levels.
The agent sees a local state $s_k^t = (\text{t}, \rho_{\text{remaining_}k})$ (current round index divided by T and remaining privacy budget divided by its initial value) and emits a continuous noise action $a_k^t = \sigma_k^t$, where σ_k^t is a value in the interval $[\sigma_{\min}, \sigma_{\max}]$. To adapt this to a fully decentralized multi-client federated setting, we apply this single-agent DNO-RL approach in a multi-agent framework.
Reward Function: The aggregated global reward signal at each round is the global model detection performance improvement (mAP) after aggregation:
$r^t = \text{mAP}(\theta_G^{t+1}) - \text{mAP}(\theta_G^t)$
This reward will encourage agents to choose noise levels that collectively enhance the detection quality, while allowing agents to set their own privacy budgets. If the raw action would breach the remaining budget constraint, each agent's raw action is projected onto the closest noise level that is achievable within the budget via an affine transformation to ensure strict adherence to the budget constraint.

4.4 Byzantine-Resilient FedAvg Aggregation

4.4.1 Byzantine Detection

The central server performs a two-stage Byzantine filtering on all client updates submitted and filters out any updates it receives more than twice from any one client. Poisoned and manipulated data to the server to influence the results of the training process, as is well known in standard FedAvg. Here, a filtering mechanism is described which addresses this threat.
The first stage uses norm-based filtering, in which the L2-norm of each client update $\|\theta_k^t - \theta^t\|_2$ is compared with a dynamic threshold τ that is based on the median of the past norms of updates. Significant deviations are indicated as possible outliers. The median is set based on the historical "norm", not a fixed value, and provides hyperparameter-free adaptation to the changing distribution of client updates.

The second stage is cosine similarity clustering; An update direction that is not aligned with the majority cluster is excluded. The two stages approach is effective as Byzantine attacks are either of significant size (which is detected by norm filtering) or of updates that are propagated in directions which go against the majority opinion (which is detected by cosine filtering) as shown experimentally.

4.4.2 Aggregation

If both the filters are passed, then passing clients are added to the Accepted_Setfor the current round . The weight average vote is then used for updating the global detection model, but only using accepted updates:

$$\theta_G^{t+1} = \frac{\sum_{k \in \text{Accepted_Set}} |D_k| \cdot \theta_k^t}{\sum_{k \in \text{Accepted_Set}} |D_k|}$$

This Byzantine-resistant FedAvg design is able to resist the poisoning of the global detection model even if up to 30% of the clients are adversarial.

4.5 SiamFC Tracking

Once detected, the positions of the confirmed drones are sent to a SiamFC (Siamese Fully Convolutional) tracker for further temporal localization over the frames. SiamFC can learn an offline similarity metric between a target template patch, extracted from the 1st frame of detection, and candidate patches in the following frames. For each new frame, the tracker calculates a response map for the cross-correlation between the template and a larger search region and positions the drone at the position of maximum cross-correlation response.
SiamFC is selected due to its real-time processing and its ability to work in the case of limited appearance cues in thermal imagery. In the context of counter-UAV applications, which have strict latency requirements showed that the fully convolutional formulation is suitable for tracking at frame rates above 58 fps and that it is an efficient, end-to-end learned tracking system. It is no longer required to fine-tune the tracker online, thus it is deployable directly after federated training of the detection model without additional per-sequence adaptation.

4.6 Evaluation strategies

System performance is evaluated using two complementary metrics:

- mAP (mean Average Precision): The standard object detection metric computed over IoU thresholds, measuring the precision-recall trade-off of the drone detector across all test sequences.
- IR Mixed Measure: A composite metric designed for thermal infrared tracking benchmark, combining success rate (overlap-based tracking accuracy) and precision score (center-location error), providing a holistic evaluation of both detection and tracking performance on Anti-UAV600 thermal sequences.

4.6.1 Overview

This chapter includes the results obtained from the evaluation of the proposed privacy preserving federated learning system for drone detection and tracking on Anti-UAV600 thermal infrared dataset activity, (ii) prevention and protection, (iii) risk mitigation, and (iv) recovery and reconstruction.It is organized around four key areas: (i) detection and activity, (ii) prevention and protection, (iii) risk mitigation and (iv) recovery and reconstruction.

The tracking accuracy against baseline methods, (ii) Byzantine client attacks such as label-flipping and gradient manipulation and (iii) privacy budget utilization efficiency per training round for ρ-zCDP and (iv) the adaptive noise behaviour learned by the DDPG agents over training rounds.

All experiments are conducted with the Anti-UAV600 thermal dataset with two different data partitioning strategies: IID and Non-IID client data partitioning. mAP is used to report results, along with the IR Mixed Measure. If applicable, results are reported as mean $\pm$ standard deviation of five independent runs where random seeds were used to obtain the results and ensure statistical reliability.

Chapter 5

Conclusion

5.1 Summary of Findings

This research advances the state of trustworthy machine learning by bridging the gap between adaptive privacy preservation and Byzantine resilience in decentralized environments. The primary contribution is the development of a Multi-Agent Reinforcement Learning (MARL) framework that transforms Differential Privacy (DP) from a static constraint into a dynamic defensive mechanism. Unlike traditional aggregation methods that often fail under the combined pressure of noise injection and adversarial poisoning, this framework introduces the Dynamic Noise Optimization (DNO-RL) method. This method intelligently calibrates Gaussian noise levels by analyzing the variance and reliability of local updates, effectively masking sensitive data while simultaneously isolating anomalous Byzantine contributions. Furthermore, by optimizing the privacy-accuracy trade-off in real-time, this work provides a scalable solution that maintains model integrity and convergence speed even under sophisticated label-flipping and gradient-manipulation attacks, offering a provable DP guarantee without the catastrophic utility loss common in existing literature.

5.2 Recommendations for Future Work

While this framework establishes a robust foundation for adaptive security, several avenues for future research remain to be explored to enhance its practical deployment. Subsequent studies should focus on extending the MARL controller to handle extreme data heterogeneity (Non-IID), as current robust aggregators still exhibit performance degradation when local data distributions vary significantly across clients. There is also a critical need to investigate the communication overhead associated with MARL training, potentially through the integration of sparse communication techniques or model compression to ensure the framework is viable for resource-constrained IoT devices. Additionally, future work could incorporate Hardware-Assisted Security, such as Trusted Execution Environments (TEEs), to provide an extra layer of protection for the MARL policy updates themselves against server-side compromise. Finally, expanding the threat model to include stealthy backdoor attacks and evaluating the system in high-stakes real-world domains, such as robotic-assisted surgery or financial fraud detection, will be essential to validating the long-term reliability and safety of adaptive privacy-preserving collaborative learning.

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